A New Clock Uses the Centers of Atoms to Keep Time

From: ScienceDaily Top Technology Mined: 2026-10-13 3 min read

Scientists in Vienna, Austria, have built the world's first self-stabilizing nuclear clock. One day, it could keep time even better than today's best atomic clocks.

Atomic clocks use **electrons**, tiny particles that move around the center of an atom. The new clock uses the **nucleus**, which is the atom's center. The **nucleus** is much less bothered by things around it.

The clock uses an element called **thorium**. Scientists can use **laser** light to switch **thorium**'s **nucleus** between two energy states.

In the new clock, a **laser** shines on a **crystal** that contains **thorium**. The **laser**'s light vibrates with a steady rhythm that can measure time. But the rhythm can drift, for example when the temperature changes. The **thorium** only soaks up the light when it is exactly right. If the **laser** drifts, the clock notices and fixes it by itself.

The team showed the clock could run steadily for more than 24 hours without help.

Word Treasure

nucleus -- The tiny center of an atom.
electrons -- Tiny particles that move around the center of an atom.
thorium -- A chemical element, which is one of the basic kinds of matter.
laser -- A very narrow, powerful beam of light.
crystal -- A solid material whose tiny parts are lined up in a neat, repeating pattern.

Background you need

Everything is made of tiny atoms, and each atom has a center called a nucleus.

The best clocks today, called atomic clocks, use the tiny vibrations of atoms to keep very exact time.

Scientists want even better clocks so they can measure things more precisely.

Break it down

WHO: Scientists in Vienna, Austria
WHAT: They built the first nuclear clock that keeps itself on time
WHERE: In a lab in Vienna, using a crystal that contains thorium
WHY: Using the nucleus of an atom could make clocks even more exact than atomic clocks

Why it matters

Super-exact clocks help scientists measure things very carefully, which can lead to new discoveries and better technology kids use.

Test what you remember

1. Where was the new clock built?

- (A) Vienna, Austria
- (B) Cairo, Egypt
- (C) Sydney, Australia
- (D) London, England

2. What part of the atom does the new clock use?

- (A) The nucleus
- (B) The electrons
- (C) The color
- (D) The outside

3. Which element does the clock use?

- (A) Silver
- (B) Oxygen
- (C) Iron
- (D) Thorium

4. What shines on the crystal in the clock?

- (A) A flashlight
- (B) The Sun
- (C) A candle
- (D) A laser

5. What can make the laser's rhythm drift?

- (A) Wind
- (B) Temperature changes
- (C) Rain
- (D) Music

6. How long did the clock run steadily without help?

- (A) Ten years
- (B) Five minutes
- (C) One second
- (D) More than 24 hours

Answer key (try the quiz first!): 1: A 2: A 3: D 4: D 5: B 6: D

Questions to think about

MY THOUGHTS (at least 20 words):